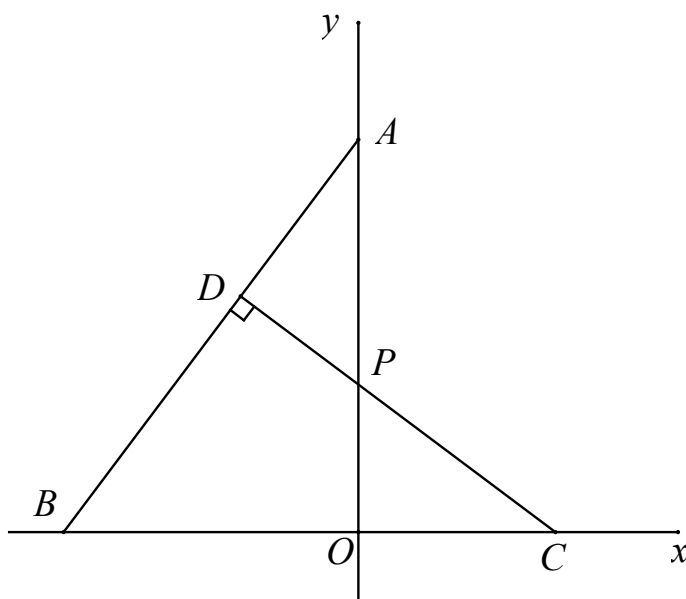


Question 1

- | | Marks |
|--|--------------|
| (a) Solve for t : $7 - 4t > 12$. | 2 |
| (b) Simplify: $\frac{x}{2} - \frac{2x+5}{6}$. | 2 |
| (c) Solve $2\sin\theta = -1$ for $0 \leq \theta \leq 2\pi$. | 2 |
| (d) Differentiate with respect to x : | |
| (i) $y = \frac{6}{\sqrt{x}}$. | 1 |
| (ii) $y = x^2 \ln x$. | 2 |
| (e) Evaluate $\int_1^5 \left(3 + \frac{2}{x}\right) dx$. | 3 |

Question 2 (START A NEW PAGE)



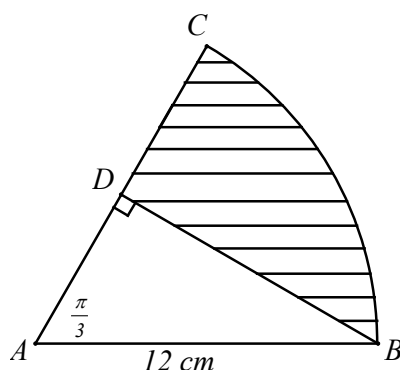
In the diagram $AB = BC$ and CD is perpendicular to AB . CD intersects the y -axis at P .

- | | |
|---|----------|
| (i) Find the length of AB . | 1 |
| (ii) Hence show that the co-ordinates of point C are $(2, 0)$. | 1 |
| (iii) Show that the equation of CD is $3x + 4y = 6$. | 3 |
| (iv) Show that the co-ordinates of P are $(0, 1\frac{1}{2})$. | 1 |
| (v) Show that the length of CP is $2\frac{1}{2}$. | 1 |
| (vi) Prove that $\triangle ADP$ is congruent to $\triangle COP$ | 3 |
| (vii) Hence calculate the area of the quadrilateral $DPOB$. | 2 |

Question 3 (START A NEW PAGE)

Marks
3

- (a) Find the equation of the tangent to the curve $y = x^3 - 4x - 1$ at the point $T(2, -1)$.
- (b)



ABC is a sector with $\angle BAC = \frac{\pi}{3}$ and $AC = AB = 12 \text{ cm}$.

- (i) Calculate the area of sector ABC . 1
- (ii) Calculate the area of the shaded region. 3
- (c) (i) The equation of a parabola is given as $y = \frac{1}{8}x^2 - x$. Rewrite the equation in the form $(x - x_o)^2 = b(y + y_o)$ where x_o , y_o and b are constants. 2
- (ii) Hence write down the focal length of the parabola and the co-ordinates of its vertex and focus. 3

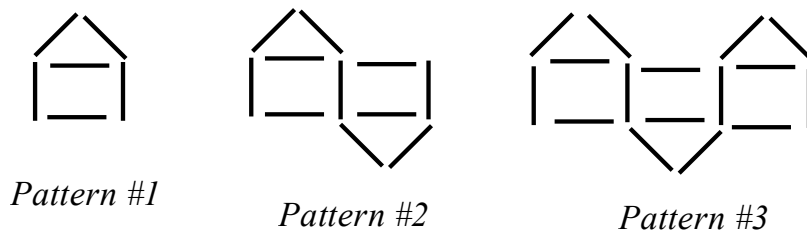
Question 4 (START A NEW PAGE)

- (a) The price of gold, $\$P$, was studied over a period of t years.
- (i) Throughout the period of study $\frac{dP}{dt} > 0$. What does this say about the price of gold? 1
- (ii) It was also observed that the rate of change of the price of gold decreased over the period of study. 1
- What does this statement imply about $\frac{d^2P}{dt^2}$?
- (b) A pool is being drained at a rate of $240t - 9600$ litres/minute.
- (i) How long does it take before the draining of the pool stops? 1
- (ii) If the pool initially held 1920000 litres of water, find the volume of water in the pool after 15 minutes. 3
- (c) In a hat are 5 red and 3 green jellybeans. Jason reaches into the hat and randomly selects two jellybeans. Find the probability that:
- (i) both selected jellybeans are red. 1
- (ii) the chosen jellybeans are different colours. 2
- (d) The tangent to the curve $y = 4\sqrt{x}$ at a point P has a slope of 2. Find the co-ordinates of point P . 3

Question 5 (START A NEW PAGE)

Marks

(a)



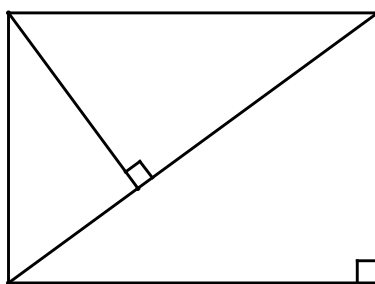
The above patterns are made using small sticks. Pattern #1 requires 6 sticks, pattern #2 requires 11 sticks and pattern #3 requires 16 sticks.

- (i) Write a formula for the number of sticks, U_n , needed to construct pattern number n . 2
 - (ii) What is the largest pattern that can be constructed from 200 sticks? 1
 - (iii) How many sticks would be needed to construct all the patterns from pattern #1 to pattern #20? 2
- (b)
- (i) Show that the curves $y = 1 + \sqrt{x}$ and $y = 7 - x$ meet at the point $(4, 3)$. 1
 - (ii) Sketch, on the same set of axes, the curves $y = 1 + \sqrt{x}$ and $y = 7 - x$ and shade the region bounded by these curves and the y -axis. 2
 - (iii) Find the volume of the solid formed when the area bounded by the $y = 1 + \sqrt{x}$, $y = 7 - x$ and the y -axis is rotated one revolution about the y -axis. 4

Question 6 (START A NEW PAGE)

- (a) (i) Show that the curves $y = \sin 2x$ and $y = \cos x$ meet at the point with abscissa $x = \frac{\pi}{6}$. 1
- (ii) Sketch, on the same set of axes, the curves $y = \sin 2x$ and $y = \cos x$ in the domain $0 \leq x \leq \frac{\pi}{2}$ and shade the region bounded only by these curves. 2
- (iii) Find the area of the shaded region described in (ii). 3

(b)



(Diagram not drawn to scale)

$ABCD$ is a rectangle with $AB = 12 \text{ cm}$, $BC = 9 \text{ cm}$ and AM perpendicular to BD .

- (i) Copy the diagram onto your answer sheet and find the length of BD . 1
- (ii) Prove that $\triangle ABM$ is similar to $\triangle BDC$. 3
- (iii) Hence find the length of BM . (Give reasons) 2

Question 7 (START A NEW PAGE)**Marks**

- (a) In an experiment, the amount of crystals, x grams, that dissolved in a solution after t minutes was given by:

$$x = 200(1 - e^{-kt})$$

- (i) After 3 minutes it was found that 50 grams of crystals had dissolved, hence find the value of k . **2**
- (ii) At what rate were the crystals dissolving after 5 minutes? Give your answer to the nearest gram/minute. **3**
- (b) Two particles A and B start from the origin at the same time and move along a straight line so that their velocities in m/s at any time t seconds are given by:

$$v_A = t^2 + 2 \text{ and } v_B = 8 - 2t$$

- (i) Show that the two particles never move with the same acceleration. **3**
- (ii) Write a formula for the position of particle A at time t seconds. **2**
- (iii) After leaving the origin, find when the two particles are again at the same position. **2**

Question 8 (START A NEW PAGE)

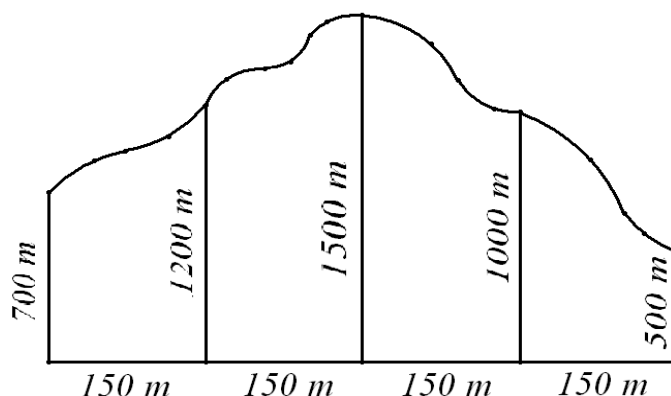
- (i) Given that $f(x) = x^2 \sqrt{10-x}$, show that $f'(x) = \frac{5x(8-x)}{2\sqrt{10-x}}$. **3**
- (ii) State the domain of the function $y = x^2 \sqrt{10-x}$. **1**
- (iii) Find all the stationary points on the curve $y = x^2 \sqrt{10-x}$ and determine their nature. **5**
- (iv) Sketch the curve $y = x^2 \sqrt{10-x}$ indicating its intercepts with the co-ordinate axes and the position of all stationary points. **3**

Question 9 (START A NEW PAGE)

Marks

- (a) Prime land along a foreshore is to be reclaimed and developed as part of a housing estate. A plan of the land to be reclaimed is shown below:

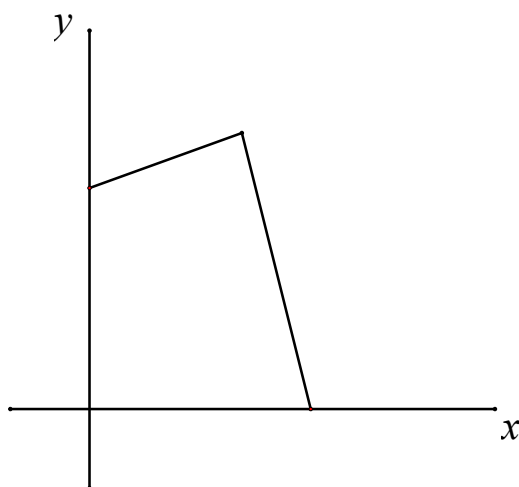
3



(Diagram not drawn to scale)

Use Simpson's rule with five function values to find the approximate area, expressed in hectares, of the land to be reclaimed.

- (b)



The point $P(x, y)$ moves so that its distance from the point $M(3, 0)$ is always twice its distance from the point $N(0, 3)$.

- | | | |
|---------|--|----------|
| (i) | Show that the equation of the locus of the point $P(x, y)$ is $x^2 + 2x + y^2 - 8y + 9 = 0$. | 2 |
| (ii) | Give a geometric description of the locus. | 2 |
| (c) (i) | Graph $y = f(x)$ if $f(x) = \begin{cases} -4x & \text{for } x < 0 \\ x^2 + 3 & \text{for } x \geq 0 \end{cases}$ | 2 |
| (ii) | Evaluate $f(-2) + f(2)$. | 1 |
| (iii) | Solve for $a > 0$: $f(a) = f(-a)$. | 2 |

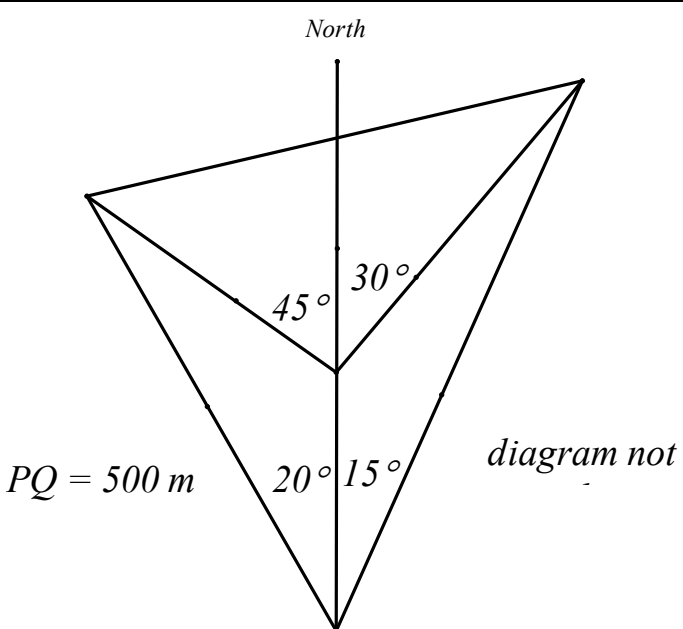
Question 10 (START A NEW PAGE)

- | | Marks |
|---|--------------|
| (a) Abigale decides to save for a car. She creates a savings account into which she deposits \$100 at the beginning of each week. The account pays interest at a rate of 0.16% per week calculated on the value of the account at the end of each week. | |
| (i) Show that Abigale's account is worth $62600(1.0016^n - 1)$ dollars at the end of n weeks. | 3 |
| (ii) Find the value of Abigale's account at the end of 52 weeks. Give your answer correct to the nearest dollar. | 1 |
| (iii) Find the number of weeks needed for Abigale's account to accumulate \$20000. (Give your answer to nearest week) | 3 |
|
(b) When a ship is travelling at a speed of v km/hr, its rate of consumption of fuel in tonnes per hour is given by $125 + 0.004v^3$. | |
| (i) Show that on a voyage of 5000km at a speed of v km/hr the formula for the total fuel used, T tonnes, is given by: | 1 |
| $T = \frac{625000}{v} + 20v^2$ | |
| (ii) Hence find the speed for the greatest fuel economy and the amount of fuel used at this speed. (Justify your answer.) | 4 |



This is the END of the examination paper



(b)	 $PQ = 500\text{ m}$ <i>diagram not</i> A tourist at point P sees two towers at X and Y on bearings of $340^\circ T$ and $015^\circ T$ respectively. The tourist then walks 500 metres due north point Q and now records the bearings as $315^\circ T$ and $030^\circ T$ respectively.	
(i)	Show that $XP = \frac{250\sqrt{2}}{\sin 25^\circ}$.	2
(ii)	Show that $YP = \frac{250\sqrt{3}}{\sin 15^\circ}$.	2
(iii)	Find the distance from tower X to tower Y correct to the nearest 10 metres.	2

Curve Sketching

	Consider the curve given by the equation: $y = 0.4x(x - 6)^2$	
(i)	Find the co-ordinates of the stationary points and determine their nature.	4
(ii)	Sketch the curve $y = 0.4x(x - 6)^2$ in the domain $-2 \leq x \leq 10$, clearly showing the intercepts with the co-ordinate axes, the stationary points and end points.	3
(iii)	What is the maximum value of $0.4x(x - 6)^2$ in the domain $-2 \leq x \leq 10$?	1

Question 1

Marks

- (a) Solve for t : $7 - 4t > 12$.

2

$$-4t > 5$$

$$t < -1\frac{1}{4}$$

- (b) Simplify: $\frac{x}{2} - \frac{2x+5}{6}$.

2

$$\begin{aligned}\frac{x}{2} - \frac{2x+5}{6} &= \frac{3x}{6} - \frac{2x+5}{6} \\ &= \frac{3x - 2x - 5}{6} \\ &= \frac{x - 5}{6}\end{aligned}$$

- (c) Solve $2\sin\theta = -1$ for $0 \leq \theta \leq 2\pi$.

2

$$\sin\theta = -\frac{1}{2}$$

$$\theta = \pi + \frac{\pi}{6} \text{ or } 2\pi - \frac{\pi}{6}$$

$$\theta = \frac{7\pi}{6} \text{ or } \frac{11\pi}{6}$$

- (d) Differentiate with respect to x :

(i) $y = \frac{6}{\sqrt{x}}$

1

$$y = 6x^{-\frac{1}{2}}$$

$$y' = -3x^{-\frac{1}{2}}$$

$$= -\frac{3}{x\sqrt{x}}$$

(ii) $y = x^2 \ln x$

2

$$y' = 2x \ln x + x^2 \cdot \frac{1}{x}$$

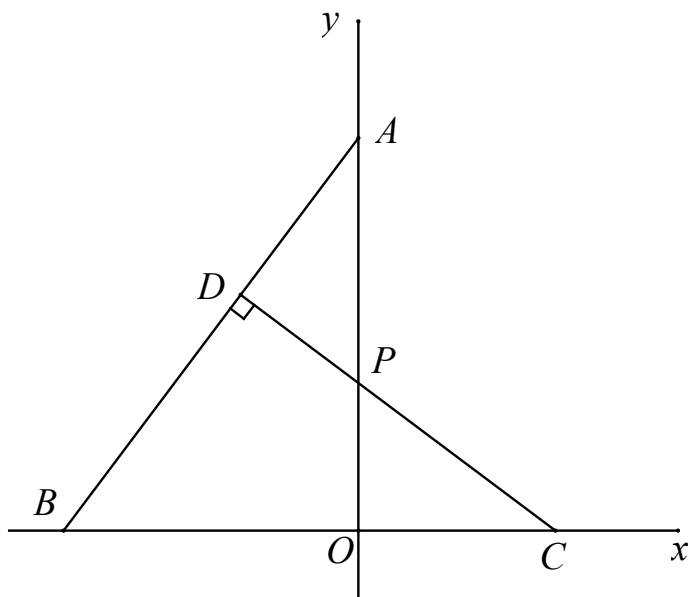
$$= 2x \ln x + x$$

- (e) Evaluate $\int_1^5 \left(3 + \frac{2}{x}\right) dx$.

3

$$\begin{aligned}\int_1^5 \left(3 + \frac{2}{x}\right) dx &= [3x + 2 \ln x]_1^5 \\ &= (15 + 2 \ln 5) - (3 + 2 \ln 1) \\ &= 12 + 2 \ln 5\end{aligned}$$

Question 2 (START A NEW PAGE)



In the diagram $AB = BC$ and CD is perpendicular to AB . CD intersects the y -axis at P .

- (i) Find the length of AB . 1

$$AB^2 = 3^2 + 4^2 \text{ (Pythagoras' Theorem)}$$

$$AB = 5$$

- (ii) Hence show that the co-ordinates of point C are $(2, 0)$. 1

$$BC = AB$$

$$\therefore BC = 5$$

$$OB + OC = 5$$

$$3 + OC = 5$$

$$OC = 2$$

$$\therefore C \text{ is } (2, 0)$$

- (iii) Show that the equation of CD is $3x + 4y = 6$. 3

$$m(AB) = \frac{4}{3}$$

$$m(CD) = -\frac{3}{4}$$

$$\text{Eqn. } CD : y - 0 = -\frac{3}{4}(x - 2)$$

$$4y = -3x + 6$$

$$3x + 4y = 6$$

- (iv) Show that the co-ordinates of P are $(0, 1\frac{1}{2})$. 1

$$3x + 4y = 6$$

$$\text{at } P \quad x = 0$$

$$\therefore 0 + 4y = 6$$

$$y = 1\frac{1}{2}$$

(v) Show that the length of CP is $2\frac{1}{2}$. 1

$$\begin{aligned} CP^2 &= \left(1\frac{1}{2}\right)^2 + 2^2 \text{ (Pythagoras' Theorem)} \\ &= 6\frac{1}{4} \\ CP &= 2\frac{1}{2} \end{aligned}$$

(vi) Prove that $\triangle ADP$ is congruent to $\triangle COP$ 3

$$\begin{aligned} &\text{In } \triangle ADP \text{ and } \triangle COP \\ \angle ADP &= \angle POC \text{ (both } 90^\circ) \\ \angle DPA &= \angle OPC \text{ (vertically opposite angles are equal)} \\ AP &= PC \text{ (both } = 2\frac{1}{2}) \\ \therefore \triangle ADP &\equiv \triangle COP \text{ (AAS)} \end{aligned}$$

(vii) Hence calculate the area of the quadrilateral $DPOB$. 2

$$\begin{aligned} \text{area } BDPO &= \text{area } \triangle BAO - \text{area } \triangle ADP \\ &= \text{area } \triangle BAO - \text{area } \triangle OPC \quad (\triangle OPC \equiv \triangle ADP) \\ &= \frac{1}{2}(3)(4) - \frac{1}{2}(2)\left(1\frac{1}{2}\right) u^2 \\ &= 4\frac{1}{2} u^2 \end{aligned}$$

Question 3 (START A NEW PAGE)

Marks
3

- (a) Find the equation of the tangent to the curve $y = x^3 - 4x - 1$ at the point $T(2, -1)$.

$$y' = 3x^2 - 4$$

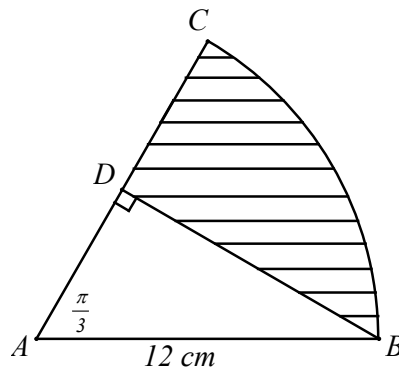
$$\begin{aligned} \text{when } x = 2, y' &= 3(2)^2 - 4 \\ &= 8 \end{aligned}$$

eqn. of tangent

$$y + 1 = 8(x - 2)$$

$$y = 8x - 17$$

- (b)



ABC is a sector with $\angle BAC = \frac{\pi}{3}$ and $AC = AB = 12 \text{ cm}$.

- (i) Calculate the area of sector ABC .

1

$$\begin{aligned} \text{area} &= \frac{1}{2}(12)^2 \cdot \left(\frac{\pi}{3}\right) \text{ cm}^2 \\ &= 24\pi \text{ cm}^2 \end{aligned}$$

- (ii) Calculate the area of the shaded region.

3

$$\frac{AD}{12} = \cos\left(\frac{\pi}{3}\right) \text{ cm}$$

$$\begin{aligned} AD &= 12 \times \frac{1}{2} \text{ cm} \\ &= 6 \text{ cm} \end{aligned}$$

$$\begin{aligned} \text{area } \triangle ADB &= \frac{1}{2} \times 6 \times 12 \times \sin\left(\frac{\pi}{3}\right) \text{ cm}^2 \\ &= 36 \times \frac{\sqrt{3}}{2} \text{ cm}^2 \\ &= 18\sqrt{3} \text{ cm}^2 \end{aligned}$$

$$\text{shaded area} = (24\pi - 18\sqrt{3}) \text{ cm}^2$$

- (c) (i) The equation of a parabola is given as $y = \frac{1}{8}x^2 - x$. Rewrite the equation in the form $(x - x_o)^2 = b(y + y_o)$ where x_o , y_o and b are constants.

2

$$y = \frac{1}{8}x^2 - x$$

$$8y = x^2 - 8x$$

$$8y + 16 = x^2 - 8x + 16$$

$$(x - 4)^2 = 8(y + 2)$$

- (ii) Hence write down the focal length of the parabola and the co-ordinates of its vertex and focus.

3

$$\text{focal length} = 2$$

$$\text{vertex } (4, -2)$$

$$\text{focus } (4, 0)$$

Question 4 (START A NEW PAGE)

(a) The price of gold, $\$P$, was studied over a period of t years.

- (i) Throughout the period of study $\frac{dP}{dt} > 0$. What does this say about the price of gold? 1

Price of gold is increasing

- (ii) It was also observed that the rate of change of the price of gold decreased over the period of study. 1

What does this statement imply about $\frac{d^2P}{dt^2}$?

$$\frac{d^2P}{dt^2} < 0$$

(b) A pool is being drained at a rate of $240t - 9600$ litres/minute.

- (i) How long does it take before the draining of the pool stops? 1

$$240t - 9600 = 0$$

$$t = \frac{9600}{240}$$

$$= 40$$

$$\text{time} = 40 \text{ minutes}$$

- (ii) If the pool initially held 1920000 litres of water, find the volume of water in the pool after 15 minutes. 3

$$\frac{dV}{dt} = 240t - 9600$$

$$V = 120t^2 - 9600t + c$$

$$\text{when } t = 0, V = 1920000$$

$$1920000 = 0 - 0 + c$$

$$c = 1920000$$

$$V = 120t^2 - 9600t + 1920000$$

$$\text{when } t = 15$$

$$V = 120(15)^2 - 9600(15) + 1920000$$

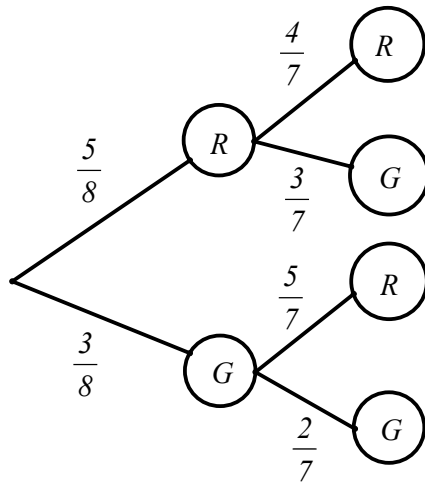
$$= 1803000$$

$$\text{volume} = 1803000 \text{ l}$$

- (c) In a hat are 5 red and 3 green jellybeans. Jason reaches into the hat and randomly selects two jellybeans. Find the probability that:

- (i) both selected jellybeans are red.

1



$$\begin{aligned} P(\text{Red, Red}) &= \frac{5}{8} \times \frac{4}{7} \\ &= \frac{5}{14} \end{aligned}$$

- (ii) the chosen jellybeans are different colours.

2

$$\begin{aligned} P(\text{different}) &= P(R, G) + P(G, R) \\ &= \frac{5}{8} \times \frac{3}{7} + \frac{3}{8} \times \frac{5}{7} \\ &= \frac{15}{28} \end{aligned}$$

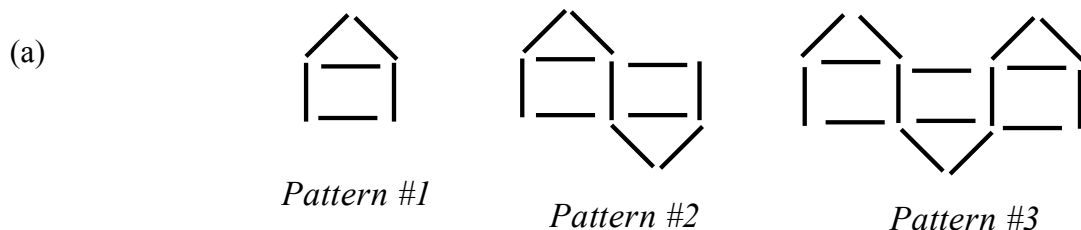
- (d) The tangent to the curve $y = 4\sqrt{x}$ at a point P has a slope of 2. Find the co-ordinates of point P .

3

$$\begin{aligned} y &= 4x^{\frac{1}{2}} \\ y' &= 2x^{-\frac{1}{2}} \\ &= \frac{2}{\sqrt{x}} \\ \therefore 2 &= \frac{2}{\sqrt{x}} \\ \sqrt{x} &= 1 \\ x &= 1 \\ y &= 4\sqrt{1} \\ y &= 4 \\ P &\text{ is } (1, 4) \end{aligned}$$

Question 5 (START A NEW PAGE)

Marks



The above patterns are made using small sticks. Pattern #1 requires 6 sticks, pattern #2 requires 11 sticks and pattern #3 requires 16 sticks.

- (i) Write a formula for the number of sticks, U_n , needed to construct pattern number n . 2

$$U_n = 6 + 5(n-1)$$

$$U_n = 5n + 1$$

- (ii) What is the largest pattern that can be constructed from 200 sticks? 1

$$5n + 1 = 200$$

$$n = 39.8$$

$\therefore$ can build pattern number 39

- (iii) How many sticks would be needed to construct all the patterns from pattern #1 to pattern #20? 2

$$S_{20} = \frac{20}{2} \{2(6) + 5(19)\}$$

$$= 1070$$

$\therefore$ need 1070 sticks

- (b) (i) Show that the curves $y = 1 + \sqrt{x}$ and $y = 7 - x$ meet at the point $(4, 3)$. 1

$$\text{when } x = 4, \quad y = 1 + \sqrt{x}$$

$$= 1 + \sqrt{4}$$

$$= 3$$

$$\text{when } x = 4, \quad y = 7 - x$$

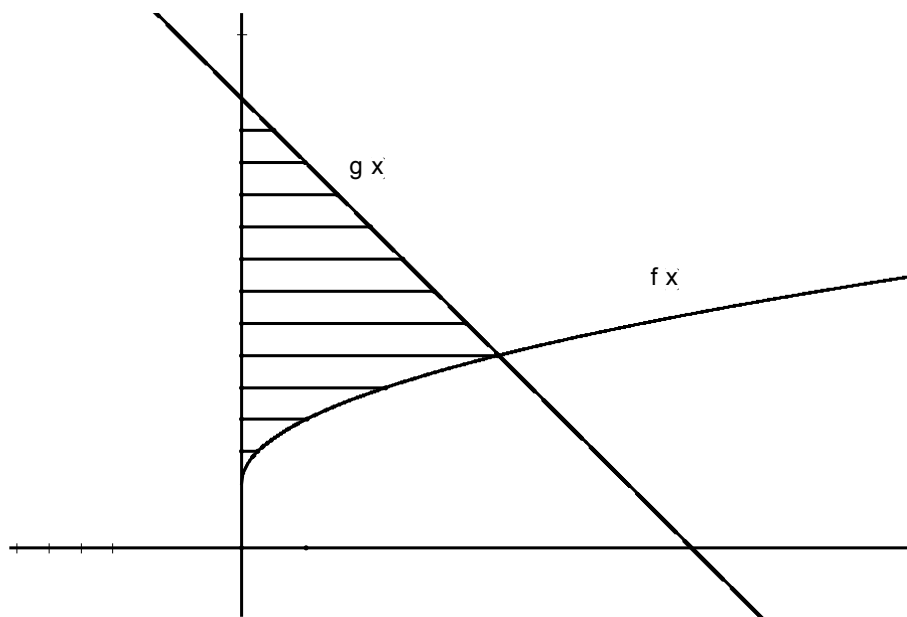
$$= 7 - 4$$

$$= 3$$

$\therefore$ point $(4, 3)$ lies on both curves

- (ii) Sketch, on the same set of axes, the curves $y = 1 + \sqrt{x}$ and $y = 7 - x$ and shade the region bounded by these curves and the y -axis.

2



- (iii) Find the volume of the solid formed when the area bounded by the $y = 1 + \sqrt{x}$, $y = 7 - x$ and the y -axis is rotated one revolution about the y -axis.

4

Volume = Vol. of revolution of $y = 1 + \sqrt{x}$ for $1 \leq y \leq 3$ + Vol. of cone for $3 \leq y \leq 7$

$$\text{Volume} = \pi \int_1^3 x^2 dy + \frac{1}{3} \pi (4)^2 (4) u^3$$

$$= \pi \int_1^3 (y-1)^4 dy + \frac{64\pi}{3} u^3$$

$$= \pi \left[\frac{(y-1)^5}{5} \right]_1^3 + \frac{64\pi}{3} u^3$$

$$= \pi \left\{ \frac{32}{5} - 0 \right\} + \frac{64\pi}{3} u^3$$

$$\text{Volume} = \frac{416\pi}{15} u^3$$

Question 6 (START A NEW PAGE)

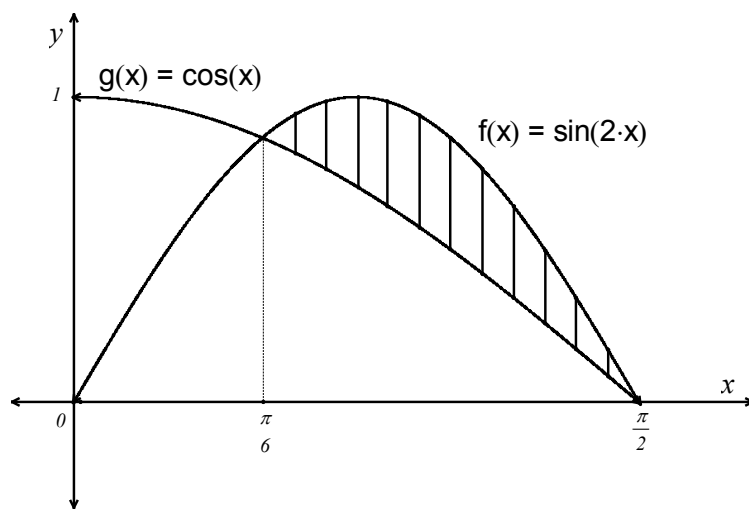
- (a) (i) Show that the curves $y = \sin 2x$ and $y = \cos x$ meet at the point with abscissa $x = \frac{\pi}{6}$. 1

$$\text{when } x = \frac{\pi}{6}, y = \sin\left(\frac{2\pi}{6}\right) \\ = \frac{\sqrt{3}}{2}$$

$$\text{when } x = \frac{\pi}{6}, y = \cos\left(\frac{\pi}{6}\right) \\ = \frac{\sqrt{3}}{2}$$

$\therefore$ point $\left(\frac{\pi}{6}, \frac{\sqrt{3}}{2}\right)$ lies on both curves

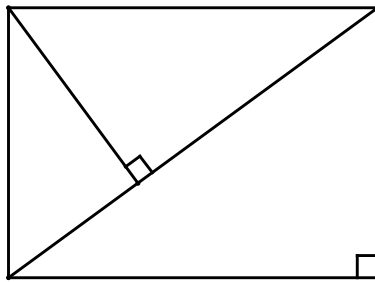
- (ii) Sketch, on the same set of axes, the curves $y = \sin 2x$ and $y = \cos x$ in the domain $0 \leq x \leq \frac{\pi}{2}$ and shade the region bounded only by these curves. 2



- (iii) Find the area of the shaded region described in (ii). 3

$$\begin{aligned} A &= \int_{\frac{\pi}{6}}^{\frac{\pi}{2}} (\sin 2x - \cos x) dx \\ &= \left[-\frac{1}{2} \cos 2x - \sin x \right]_{\frac{\pi}{6}}^{\frac{\pi}{2}} \\ &= \left(-\frac{1}{2} \cos \pi - \sin \frac{\pi}{2} \right) - \left(-\frac{1}{2} \cos \frac{\pi}{3} - \sin \frac{\pi}{6} \right) \\ &= \left(\frac{1}{2} - 1 \right) - \left(-\frac{1}{4} - \frac{1}{2} \right) \\ &= \frac{1}{4} \\ \text{Area} &= \frac{1}{4} u^2 \end{aligned}$$

(b)

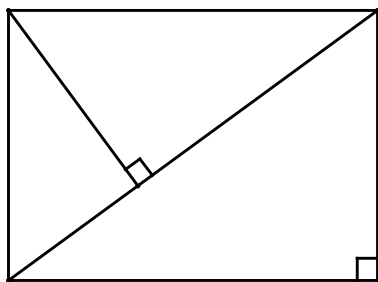


(Diagram not drawn to scale)

$ABCD$ is a rectangle with $AB = 12\text{ cm}$, $BC = 9\text{ cm}$ and AM perpendicular to BD .

- (i) Copy the diagram onto your answer sheet and find the length of BD .

1



(Diagram not drawn to scale)

$$\begin{aligned} BD^2 &= DC^2 + CB^2 \quad (\text{Pythagoras' Theorem}) \\ &= 12^2 + 9^2 \\ &= 225 \\ BD &= 15\text{ cm} \end{aligned}$$

- (ii) Prove that $\triangle ABM$ is similar to $\triangle BDC$.

3

$AB \parallel DC$ (opposite sides of a rectangle are parallel)

In $\triangle ABM$ and $\triangle BDC$

$\angle ABM = \angle BDC$ (alternate angles are equal as $AB \parallel DC$)

$\angle AMB = \angle DCB$ (both 90°)

$\triangle ABM \parallel \triangle BDC$ (equiangular)

- (iii) Hence find the length of BM . (Give reasons)

2

$$\begin{aligned} \frac{BM}{12} &= \frac{12}{15} \quad (\text{ratios of corresponding sides are equal in similar triangles}) \\ BM &= 9.6\text{ cm} \end{aligned}$$

Question 7 (START A NEW PAGE)

Marks

- (a) In an experiment, the amount of crystals, x grams, that dissolved in a solution after t minutes was given by:

$$x = 200(1 - e^{-kt})$$

- (i) After 3 minutes it was found that 50 grams of crystals had dissolved, hence find the value of k .

2

when $t = 3$, $x = 50$

$$50 = 200(1 - e^{-3k})$$

$$0.25 = 1 - e^{-3k}$$

$$e^{-3k} = 0.75$$

$$-3k = \ln(0.75)$$

$$k = -\frac{\ln(0.75)}{3} \quad (\approx 0.09589)$$

- (ii) At what rate were the crystals dissolving after 5 minutes? Give your answer to the nearest gram/minute.

3

$$x = 200(1 - e^{-kt})$$

$$\frac{dx}{dt} = 200ke^{-kt} \text{ where } k = -\frac{\ln(0.75)}{3}$$

when $t = 5$

$$\frac{dx}{dt} = 200\left(-\frac{\ln(0.75)}{3}\right)e^{5\left(\frac{\ln(0.75)}{3}\right)}$$

$$\approx 30.9779566$$

rate = 31 grams/minute

- (b) Two particles A and B start from the origin at the same time and move along a straight line so that their velocities in m/s at any time t seconds are given by:

$$v_A = t^2 + 2 \text{ and } v_B = 8 - 2t$$

- (i) Show that the two particles never move with the same acceleration.

3

$$a_A = \frac{dv_A}{dt}$$

$$= 2t$$

$$a_B = \frac{dv_B}{dt}$$

$$= -2$$

$$\text{if } a_A = a_B$$

$$2t = -2$$

$$t = -1$$

$$\text{but } t \geq 0$$

$\therefore$ no solution

$\therefore$ particles never move with the same acceleration

- (ii) Write a formula for the position of particle A at time t seconds.

2

$$x_A = \frac{1}{3}t^3 + 2t + c$$

$$\text{when } t = 0, x_A = 0 \Rightarrow c = 0$$

$$x_A = \frac{1}{3}t^3 + 2t$$

- (iii) After leaving the origin, find when the two particles are again at the same position.

2

$$x_B = 8t - t^2 + d$$

$$\text{when } t = 0, x_B = 0 \Rightarrow d = 0$$

$$x_B = 8t - t^2$$

$$\text{when } x_A = x_B$$

$$\frac{1}{3}t^3 + 2t = 8t - t^2$$

$$t^3 + 3t^2 - 18t = 0$$

$$t(t+6)(t-3) = 0$$

$$t = 3 \quad (t > 0)$$

$\therefore$ at same position after 3 seconds

Question 8 (START A NEW PAGE)

- (i) Given that $f(x) = x^2 \sqrt{10-x}$, show that $f'(x) = \frac{5x(8-x)}{2\sqrt{10-x}}$.

3

$$\begin{aligned} f(x) &= x^2 (10-x)^{\frac{1}{2}} \\ &= (2x) \left\{ (10-x)^{\frac{1}{2}} \right\} + (x^2) \left\{ \frac{1}{2} (10-x)^{-\frac{1}{2}} \times -1 \right\} \\ &= 2x\sqrt{10-x} - \frac{x^2}{2\sqrt{10-x}} \\ &= \frac{4x(10-x) - x^2}{2\sqrt{10-x}} \\ &= \frac{40x - 5x^2}{2\sqrt{10-x}} \\ &= \frac{5x(8-x)}{2\sqrt{10-x}} \end{aligned}$$

- (ii) State the domain of the function $y = x^2 \sqrt{10-x}$.

1

$$\begin{aligned} \text{Domain: } 10-x &\geq 0 \\ \therefore x &\leq 10 \end{aligned}$$

- (iii) Find all the stationary points on the curve $y = x^2 \sqrt{10-x}$ and determine their nature.

5

$$\text{for stat. pts } \frac{dy}{dx} = 0$$

$$\therefore \frac{5x(8-x)}{2\sqrt{10-x}} = 0$$

$$x = 0 \text{ or } 8$$

$$\begin{aligned} x = 0, y = 0 \text{ and } x = 8, y &= 8^2 \sqrt{10-8} \\ &= 64\sqrt{2} \end{aligned}$$

$$\text{stat. pts. are } (0,0) \text{ and } (8, 64\sqrt{2})$$

Test nature

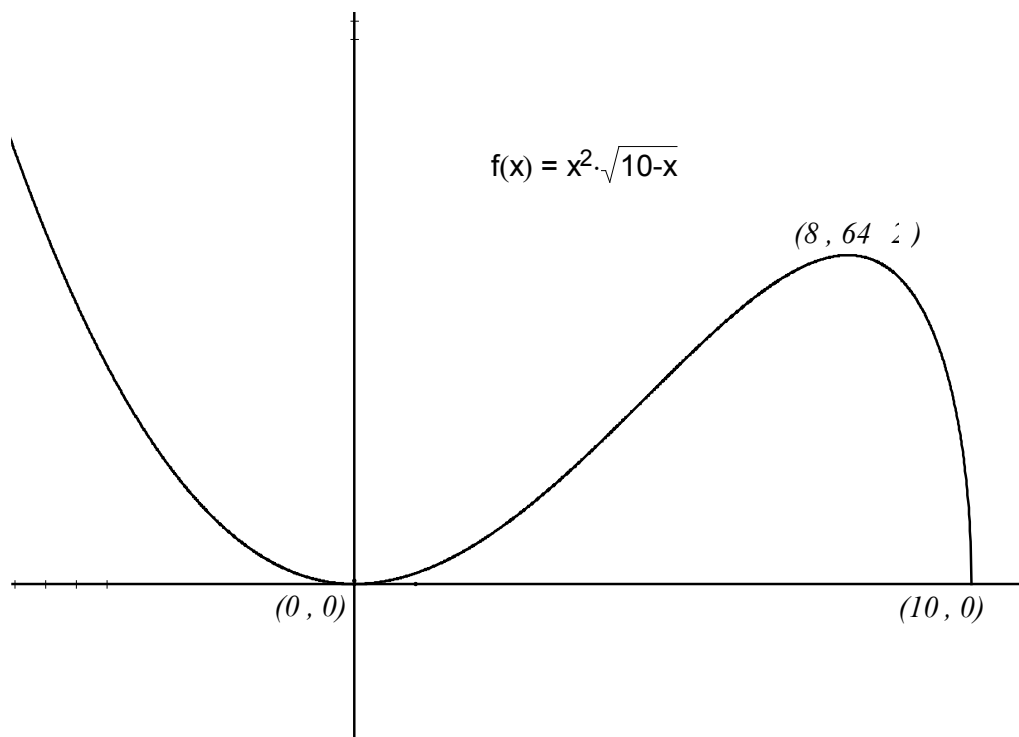
x	-1	0	1
$\frac{dy}{dx}$	$-\frac{45}{2\sqrt{11}}$	0	$\frac{35}{6}$
	< 0		> 0
	\ _ / \text{ local min.}		

x	7	8	9
$\frac{dy}{dx}$	$\frac{35}{2\sqrt{3}}$	0	$-\frac{45}{2}$
	> 0		< 0
	/ _ \text{ local max.}		

$$\therefore \text{local min. tp. at } (0,0) \text{ and local max. tp. at } (8, 64\sqrt{2})$$

- (iv) Sketch the curve $y = x^2\sqrt{10-x}$ indicating its intercepts with the co-ordinate axes and the position of all stationary points.

3

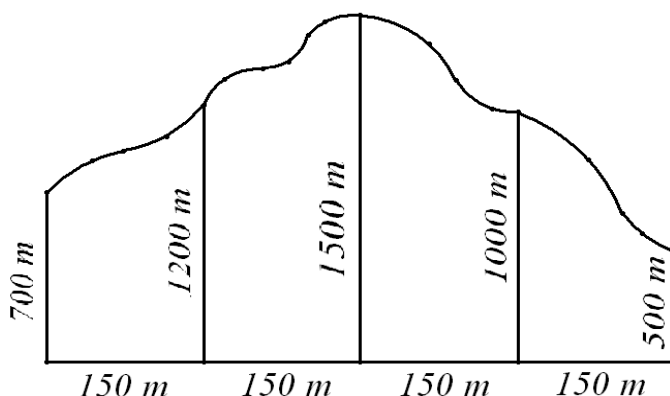


Question 9 (START A NEW PAGE)

Marks

- (a) Prime land along a foreshore is to be reclaimed and developed as part of a housing estate. A plan of the land to be reclaimed is shown below:

3



(Diagram not drawn to scale)

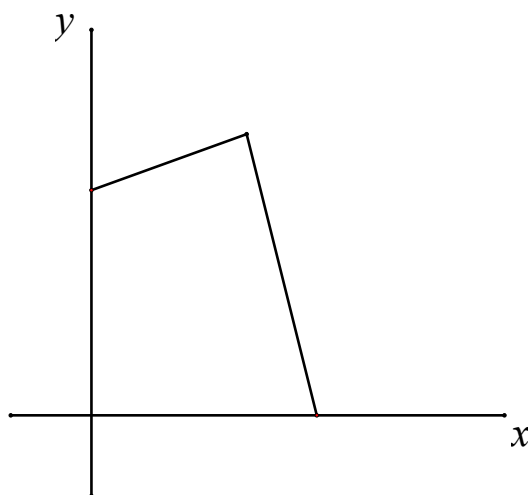
Use Simpson's rule with five function values to find the approximate area, expressed in hectares, of the land to be reclaimed.

$$\text{area} \approx \frac{300}{6}(700 + 4 \times 1200 + 1500) + \frac{300}{6}(1500 + 4 \times 1000 + 500) \text{ m}^2$$

$$\approx 650000 \text{ m}^2$$

$$\text{area} \approx 65 \text{ ha}$$

- (b)



The point $P(x, y)$ moves so that its distance from the point $M(3, 0)$ is always twice its distance from the point $N(0, 3)$.

- (i) Show that the equation of the locus of the point $P(x, y)$ is $x^2 + 2x + y^2 - 8y + 9 = 0$. 2

$$PM = 2PN$$

$$PM^2 = 4PN^2$$

$$(x-3)^2 + y^2 = 4\{x^2 + (y-3)^2\}$$

$$x^2 - 6x + 9 + y^2 = 4(x^2 + y^2 - 6y + 9)$$

$$x^2 - 6x + 9 + y^2 = 4x^2 + 4y^2 - 24y + 36$$

$$3x^2 + 6x + 3y^2 - 24y + 27 = 0$$

$$x^2 + 2x + y^2 - 8y + 9 = 0$$

- (ii) Give a geometric description of the locus. 2

$$x^2 + 2x + y^2 - 8y + 9 = 0$$

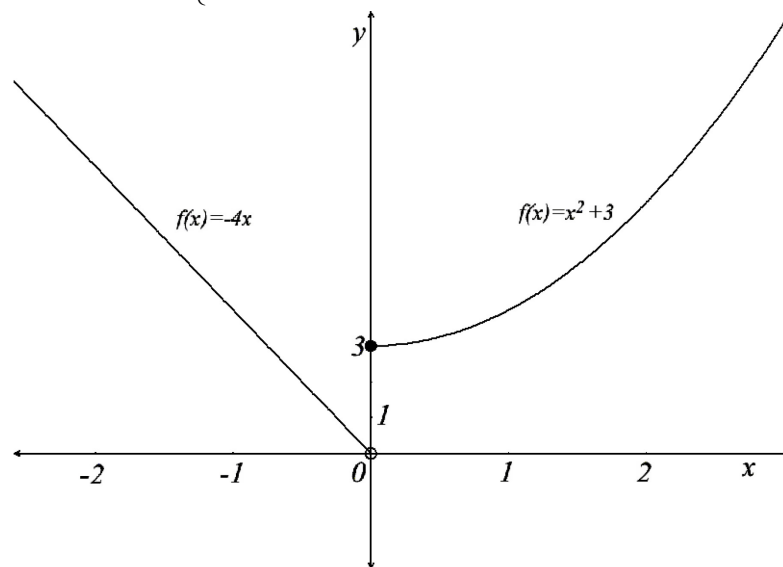
$$x^2 + 2x + y^2 - 8y = -9$$

$$x^2 + 2x + 1 + y^2 - 8y + 16 = -9 + 1 + 16$$

$$(x+1)^2 + (y-4)^2 = 8$$

locus is a circle with centre at $(-1, 4)$ and radius $= 2\sqrt{2}$

- (c) (i) Graph $y = f(x)$ if $f(x) = \begin{cases} -4x & \text{for } x < 0 \\ x^2 + 3 & \text{for } x \geq 0 \end{cases}$ 2



- (ii) Evaluate $f(-2) + f(2)$. 1

$$f(-2) + f(2) = \{-4(-2)\} + \{(2)^2 + 3\}$$

$$= 8 + 7$$

$$= 15$$

- (iii) Solve for $a > 0 : f(a) = f(-a)$. 2

$$a^2 + 3 = 4a$$

$$a^2 - 4a + 3 = 0$$

$$(a-1)(a-3) = 0$$

$$a = 1 \text{ or } 3$$

Question 10 (START A NEW PAGE)**Marks**

- (a) Abigale decides to save for a car. She creates a savings account into which she deposits \$100 at the beginning of each week. The account pays interest at a rate of 0.16% per week calculated on the value of the account at the end of each week.

- (i) Show that Abigale's account is worth $62600(1.0016^n - 1)$ dollars at the end of n weeks.

3

Let A_n dollars be the value of Abigale's account at the end of n weeks

$$A_1 = 100 \times 1.0016$$

$$A_2 = (A_1 + 100) \times 1.0016$$

$$= 100 \times 1.0016^2 + 100 \times 1.0016$$

$$= 100(1.0016^2 + 1.0016)$$

$$A_3 = (A_2 + 100) \times 1.0016$$

$$= 100(1.0016^3 + 1.0016^2 + 1.0016)$$

$$A_n = 100(1.0016^n + 1.0016^{n-1} + \dots + 1.0016^2 + 1.0016)$$

$$= 100 \times \frac{1.0016(1.0016^n - 1)}{1.0016 - 1}$$

$$= \frac{100.16(1.0016^n - 1)}{0.0016}$$

$$A_n = 62600(1.0016^n - 1)$$

- (ii) Find the value of Abigale's account at the end of 52 weeks. Give your answer correct to the nearest dollar.

1

$$A_n = 62600(1.0016^n - 1)$$

when $n = 52$

$$A_{52} = 62600(1.0016^{52} - 1)$$

$$\approx 5426.60$$

Value of account = \$5427 (to the nearest dollar)

- (iii) Find the number of weeks needed for Abigale's account to accumulate \$20000.
(Give your answer to nearest week)

$$20000 = 62600(1.0016^n - 1)$$

$$1.0016^n - 1 = \frac{20000}{62600}$$

$$1.0016^n = \frac{413}{313}$$

$$n \ln(1.0016) = \ln\left(\frac{413}{313}\right)$$

$$n = \frac{\ln\left(\frac{413}{313}\right)}{\ln(1.0016)}$$

$$\approx 173.416$$

$$\therefore \text{no. week} = 173$$

(Note : 174 weeks is possibly a more correct answer since at 173 weeks the account would be slightly less than \$20000)

- (b) When a ship is travelling at a speed of v km/hr, its rate of consumption of fuel in tonnes per hour is given by $125 + 0.004v^3$.

- (i) Show that on a voyage of 5000km at a speed of v km/hr the formula for the total fuel used, T tonnes, is given by:

$$T = \frac{625000}{v} + 20v^2$$

$$\text{time} = \frac{5000}{v}$$

Amt. of fuel = time $\times$ consumption rate

$$T = \frac{5000}{v} \times (125 + 0.004v^3)$$

$$T = \frac{625000}{v} + 20v^2$$

- (ii) Hence find the speed for the greatest fuel economy and the amount of fuel used at this speed. (Justify your answer.)

$$T = \frac{625000}{v} + 20v^2$$

$$\frac{dT}{dv} = -\frac{625000}{v^2} + 40v$$

$$\text{for stat. pt. } \frac{dT}{dv} = 0$$

$$-\frac{625000}{v^2} + 40v = 0$$

$$40v = \frac{625000}{v^2}$$

$$v^3 = 15625$$

$$v = 25$$

$$\frac{d^2T}{dv^2} = \frac{1250000}{v^3} + 40$$

$$\text{when } v = 25, \quad \frac{d^2T}{dv^2} = \frac{1250000}{25^3} + 40 > 0$$

$\therefore$ concave up, $\therefore$ local min. tp.

Since the function is continuous for $v > 0$ and there is only one stat. pt. then the local min. tp. is the absolute minimum.

$\therefore$ minimum speed = 25 km/hr and amount fuel used = 37500 tonnes



This is the END of the examination paper

